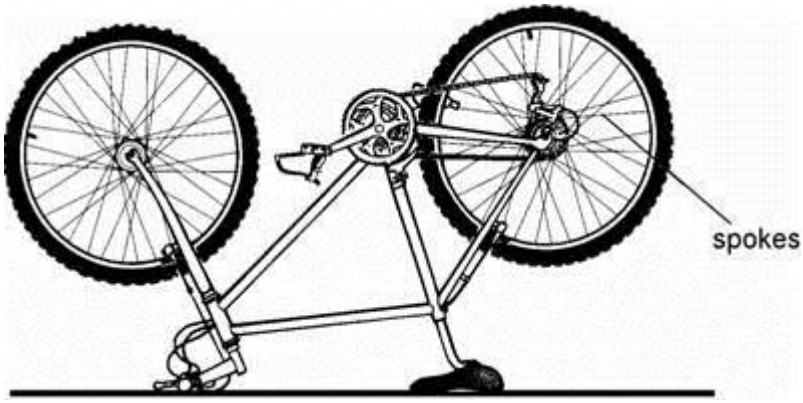
Q1.

Sam turns his bicycle upside down and turns the pedals fast so that the back wheel spins.



Sam holds a piece of card close to the back wheel.
The spokes hit the piece of card and Sam hears a 'musical' tone.

(a) Explain why the card produces a 'musical' tone.

.....

.....

1 mark

(b) When the spokes hit the piece of card, the wheel slows down

(i) Describe the energy change which occurs as the wheel slows down.

.....

.....

1 mark

(ii) Describe and explain what will happen to the pitch of the 'musical' tone as the wheel slows down.

.....

.....

2 marks
Maximum 4 marks

Q1.

(a) the spokes make it vibrate

accept 'it vibrates'

accept 'the spokes hit it many times each second'

1

(b) (i) kinetic energy in the wheel is transferred to the surroundings by sound **or** heat

*accept 'K.E. to sound **or** heat'*
***or** 'from the wheel to the surroundings'*

1

(ii) it will get lower

*accept 'it will get flatter **or** deeper'*

1

because the spokes do not hit the card so fast **or** so often **or** so frequently

accept 'because there is a lower frequency of vibration'

1

[4]